Mathematical Manuscripts
Overview and Verification Guide

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Abstract

Mathematical manuscripts contributing to an ongoing Wave Confinement Theory (WCT) research program. Formal verification is used for rigorous development and correction, not abandonment of foundational physical objectives. Analytic arguments, Lean declarations, numerical evidence and physical interpretations remain distinct. External peer review is not implied.

Research purpose and interpretation

The objective is to investigate whether confinement, resonance, curvature, phase dynamics and nonlinear fields can support a mathematically consistent, experimentally testable physical theory. Independent mathematical results do not themselves validate WCT physics; obstructions only exclude their precisely quantified constructions. Failed routes should lead to explicit new mathematical or empirical tests, not assumed repairs or a compulsory engineering-only pivot. Consistent ghost-free parent models and viable relativistic structure remain separate research obligations. See https://github.com/rickyjreyes/math/blob/main/RESEARCH_DIRECTION.md.

Flagship: fourth-order cubic–quintic variational PDE

On $H^2(\mathbb{R}^2;\mathbb{C})$, for $k,\beta,\gamma>0$, consider

$$H(u)=\|(\Delta+k^2)u\|_2^2-\frac{\beta}{2}\|u\|_4^4+\frac{\gamma}{3}\|u\|_6^6.$$

The primary family presents sufficient conditions for strict subadditivity, minimizer existence and orbital stability in the complete minimizer set. These claims concern a declared independent mathematical Hamiltonian.

Result families

Family/ID	Contents
F01	Strict binding, minimizers, orbital stability and negative trials
F02	Higher-derivative constraint counting and Sobolev UV obstruction
F03	Spectral–phase exact identity and closed-curve counterexample
F04	Nonlinear finite Fourier subspace non-invariance

F05	Positive curvature whole-space nonattainment
F06	Finite-wavenumber dispersion minimum
F07	Exact weighted phase-locking variational result
F08	Rank and signature constraints on scalar-gradient tensors
F09	11-by-31 structural local identifiability obstruction
F10	No L^2 state supported on a single Fourier shell
F11	E13/E14 band-pass dissipation; E16 linear spectral selection; E59 fixed-annulus projection; E62 Fourier fraction
E61	Centered continuum Swift–Hohenberg spectral threshold
M6B	Nonuniqueness counterexamples for curvature closure and fixed-mass inverse map
F12	E26 conditional historical additive-denominator L^2 estimate
WCT E68	Localized-energy counterexample and corrected L^n, H^2 estimates

Regularized curvature scope. E26 treats the historical additive-denominator expression and assumes a uniform positive lower bound on its modulus. The separately catalogued M2/E17 result treats a corrected modulus-squared, phase-covariant regularization; the two operators should not be conflated.

Independent reproduction

Browse the catalogue and PDFs at https://github.com/rickyjreyes/math/blob/main/CONTENTS.md. Lean declarations and explicit verification status: https://github.com/rickyjreyes/math/blob/main/lean/formalization.yaml. The maintained Lean project includes formalizations of F01 nonlinear-PDE manuscript results. Exact scope and the remaining analytic-only manuscripts are recorded in the manifest; mathematical proof alone does not validate physical interpretation.